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Lunar highland rocks: Element partitioning among minerals II: Electron microprobe analyses of Al, P, Ca, Ti, Cr, Mn and Fe in olivine

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Abstract—Electron probe measurements are presented for Al, P, Ca, Ti, Cr and Mn in lunar olivines from anorthosites, granulitic impactites, and rocks in the *mg*-rich plutonic trend. The FeO/MnO ratio for lunar olivines lies between 80 and 110 with little difference among the rock types. Comparison with terrestrial rocks and meteorites is complicated by inter-mineral partitioning of FeO/MnO: thus for terrestrial peridotites and eclogites from the earth's mantle, the FeO/MnO ranges are—olivine 60–120, mostly near 80, orthopyroxene 30–60, clinopyroxene 20–90, garnet 15–70. However, for simple crystal-liquid partitioning models, the bulk moon should have a higher FeO/MnO ratio (100 ± 10?) than parent bodies for most meteorites. The low values of Ca in lunar olivines indicate slow cooling to subsolidus temperatures, with blocking temperatures of ~750 and ~1000°C for 67667 and 60255,73α from Finnerty-Boyd experiments. The olivine from 15445,60 has the lowest Ca, and an empirical model suggests very tentatively that the blocking temperature might be very low (400°C). Experimental calibration of Ca exchange between olivine and plagioclase is needed. Titanium is very low in Fe-rich olivines from anorthosites even though both Ti and Fe tend to become enriched in liquid during fractional crystallization: this is an important paradox. Chromium correlates weakly with Mg in olivine, and a good correlation between Ca and Cr suggests that Cr migrates out of olivine during subsolidus cooling. Except for Ca and Mn, olivine from anorthosites has consistently lower values of minor elements than for other rock types, and formation from a chemically distinct system is implied.

INTRODUCTION

This is the second part of a series on major, minor and trace elements in minerals of feldspar-rich lunar rocks. Part I (Hansen *et al.*, 1979) gave electron microprobe analyses of Na, Mg, K and Fe in plagioclase, and an accompanying paper (Steele *et al.*, 1981) gives ion microprobe analyses of Li, Na, Mg, K, Ti, Sr and Ba in plagioclase. This present contribution is devoted to electron microprobe analyses of olivine, and expands on an abstract (Hansen *et al.*, 1980), and a paper (Steele and Smith, 1975). Emphasis is placed on pristine highlands rocks from anorthositic and *mg*-rich plutonic trends (Bickel and Warner, 1978; Warren and Wasson, 1978, 1979), and on granulitic impactites lying between the two trends (Warner *et al.*, 1977). A summary of the petrology and geochemistry of pristine highlands rocks was given by Norman and Ryder (1979) and of the components of highland breccias by Ryder (1979).

Table 1. Electron microprobe analyses of olivines.

Sample	Fo mole %	FeO wt.%	Al_2O_3 ppm	P_2O_5 ppm	CaO ppm	TiO_2 ppm	Cr_2O_3 ppm	MnO ppm	Number of grains
granulitic impactite									
15304,59	72	25.4	230–1150(550)	60–330(240)	d	390–540(470)	90–260(150)	2280–2400(2330)	4
67746,1	75	23.0	300–360(310)	180–220(190)	490–710(590)	360–610(440)	60–270(180)	2110–2160(2140)	4
73215,234	76	19.5–24.4	710–1070(920)	340–600(460)	d	880–2230(1890)	730–2040(1200)	1880–1990(1930)	3
79215,62	73	24.6	220–550(350)	50–90(70)	710	220–440(350)	330–490(430)	2380–2740(2580)	5
mg-rich plutonic rock or clast									
15364,2α	71	26.0	140–210(190)	70–250(150)	520–860(690)	310–380(340)	10–140(70)	2490–2730(2620)	4
73215,234	91	8.8	320–400(350)	30–130(80)	420–510(470)	480–770(590)	180–350(290)	870–930(910)	5
76255,72	88	11.6	130–190(160)	250–750(600)	670–880(770)	70–200(110)	290–390(360)	1070–1190(1150)	6
76535,49	87	12.5	80–240(170)	30–260(140)	180–250(220)	0–310(190)	70–190(130)	1290–1560(1390)	13
77135,121	77	21.3	280–370(330)	20–50(40)	730–1210(710)	390–910(540)	440–780(640)	2340–2520(2420)	5
anorthosite									
15437,7	66	25.3–30.7	110–180(160)	0–40(30)	390–480(430)	0–300(190)	0–70(30)	2010–3780(3340)	5
67075,51	59	35.4	130–180(150)	10–70–(40)	520–780(650)	60–140(90)	0–70(30)	3400–4060(3690)	7
67075,43	45	45.1	190(190)	50,70(60)	d	50,100(80)	0,50(30)	4560,5640(5100)	2
67455,106	51	41.0	100–590(240)	70–180(130)	330–560(400)	0–280(150)	10–140(70)	4520–4730(4580)	4
67635,2β	61	33.9	390,400(400)	20,70(50)	d	30,60(50)	0	3980,4070(4020)	2
67637,1β	60	34.6	140–320(220)	10–90(30)	630–670(650)	50–110(70)	0–40(10)	3760–4480(4270)	6
miscellaneous									
60619,2[a]	72	25.4	280–360(330)	60–190(120)	d	360–600(530)	200–360(260)	3020–3150(3060)	5
60255,73β[b]	74	23.8	300–420(360)	210–930(680)	1590–1820(1670)	270–410(370)	1290–2410(1690)	2360–2550(2460)	4
67667,1γ[c]	71	23.9–28.1	110–240(170)	40–170(90)	370–730(540)	390–660(490)	80–180(130)	2390–2810(2590)	9
detection level			*40*	*40*	*40*	*40*	*40*	*60*	

[a] granoblastic anorthosite, [b] feldspathic microgabbro, [c] feldspathic lherzolite, [d] grains too small to avoid fluorescence.

ANALYTICAL PROCEDURE

Routine analyses of major elements were made on an ARL-EMX-SM electron microprobe with an EDS system using a data reduction system like that of Reed and Ware (1975).

Accurate analysis of minor elements (Table 1) were made with an automated spectrometer system using a focused beam of 2–10 μm diameter at 25 kev and 2 μamp beam current. Reference standards were: Al, $Di_{85}Jd_{15}$ glass; P, $Ca_2P_2O_7$ crystal; Ca, $Di_{85}Jd_{15}$ glass; Ti, Corning V glass, 0.48 wt.%; Cr, Corning V glass, 0.52 wt.%; Mn, Corning W glass, 0.50 wt.%; Fe, P140 olivine; Ni, Corning X glass, 0.56 wt.%; Na, $Di_{85}Jd_{15}$ glass.

Background profiles were determined for each element in olivine with a typical major element composition by step scanning the spectrometer across the peak position. Count times of up to 60 seconds were used for each step, up to 25 steps were measured, count rates were accumulated in a computer, and results were plotted automatically. Using these profiles, background positions were chosen on either side of the peak and a linear variation in background was confirmed by eye-estimate of the plotted points. Based on the observed background count rates, peak and background counting times were calculated to achieve detection levels with 95% confidence ($\pm 2\sigma$) according to Reed (1973). These detection limits are: P_2O_5 40 ppm, TiO_2 40 ppm, Al_2O_3 40 ppm, Cr_2O_3 40 ppm, MnO 60 ppm, NiO 100 ppm, CaO 40 ppm, Na_2O 30 ppm. Data for both major and minor elements for both peak and background positions were obtained for each analysis point. Although major element count rates were affected seriously by dead-time, derived concentrations were sufficient to allow standard ZAF matrix corrections for minor and trace elements. Secondary fluorescence should be trivial for all elements except Ca because analyses were made at sufficient distance from grains containing high concentration of the element being analysed. For Ca, data are given only for olivine grains larger than 50 μm, for which secondary fluorescence from adjacent plagioclase should be insignificant. All olivine grains were smaller than 50 μm in 15304, 60619, 67635 and the granulitic impact clast in 73215. For 79215, only one grain was larger than 50 μm. The range of analyses is greater than the 3σ counting error for some elements in some specimens, as seen from the ranges in Table 1. Most values of MgO and FeO differed only by counting precision, and only the mean value of FeO and mol.% Fo is quoted. No attempt was made to relate chemical variation to the texture but a detailed study should be made to test whether grains show concentric zoning.

DISCUSSION OF INDIVIDUAL ELEMENTS

Manganese

All olivine data show a good correlation between MnO and FeO (Fig. 1), with most values of FeO/MnO between 80 and 110. Ratios for granulitic impactites tend to be higher than those for rocks in the *mg*-rich plutonic group while all anorthosites have FeO/MnO near 90.

Although the FeO/MnO ratio has a general cosmochemical significance because Mn should condense later than Fe during equilibrium progressive condensation of the solar nebula with C/O ~ 0.6 (Grossman and Larimer, 1974), detailed interpretation of the FeO/MnO ratio must take into account inter-mineral partitioning during crystal-liquid differentiation. For peridotites and eclogites from the Earth's mantle (Delaney *et al.,* 1979a), the FeO/MnO ratios (Fig. 2) range as follows: olivine 60–120, mostly near 80; orthopyroxene 30–60, mostly near 50; clinopyroxene 20–90, garnet 15–70. Further data for Mn in terrestrial olivines were given by Simkin and Smith (1970, Fig. 1). The large ranges for clinopyroxene and garnet result partly from a temperature-controlled exchange reaction (Dela-

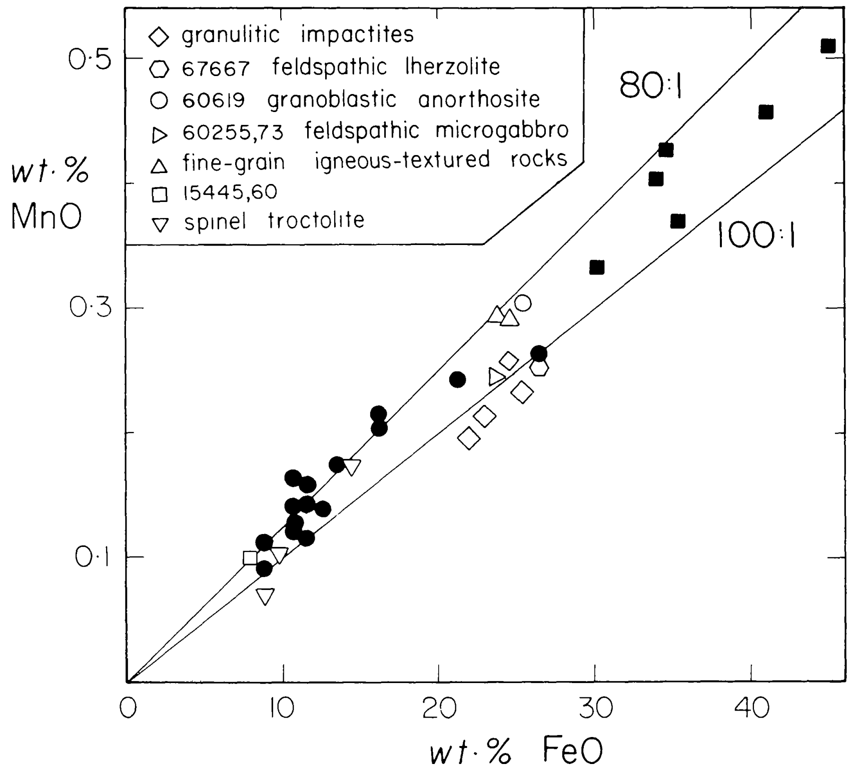

Fig. 1. Relationship between MnO and FeO for olivines from lunar non-mare rocks. Data from Steele and Smith (1975) and this paper (Table 1). Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols on figure.

ney *et al.*, 1979b). For a series of alkaline volcanic rocks, Guérin *et al.* (1979) found that Fe/Mn ranged from ~100–60 at the beginning of differentiation to ~14–10 at the end, and that Fe/Mn was lower for pyroxene than for both olivine and magnetite in each rock. Duke (1976) found that for lbar and 1125–1250°C under anhydrous conditions FeO/MnO was higher by 1.2 times on average for olivine than for clinopyroxene. For the system MgO-CaO-Na_2O-Al_2O_3-SiO_2 at 1 atmosphere and 1250–1450°C, Watson (1979) found a strong temperature dependence on Mn distribution but with a tendency for Mn to favor basic melt over olivine, and olivine over acidic melt. For seven basaltic compositions at 1 atmosphere and 1100–1500°C (Takahashi, 1978) the FeO/MnO ratio was similar for coexisting olivine and liquid but with a tendency to be higher for the olivine by ~20%. Other data are summarized in Irving (1978).

It is obvious that the FeO/MnO ratio cannot be used as a simple test of the relation between the chemical compositions of the earth and moon. The strong partitioning between olivine, garnet and pyroxenes in peridotite xenoliths from the earth (Fig. 2) is a warning that the FeO/MnO ratio might change considerably with depth as high-pressure mineral transformations occur. Taken at face value, the range of FeO/MnO for olivine in mantle peridotites from the earth (60–120) encompasses that for olivines from the moon (mainly 80–100), and it might be argued therefrom that the earth and moon inherited their Fe/Mn ratio from the same reservoir. Indeed, it might be argued that this supports origin of the moon by fission from the earth! For a pyrolite composition, the equivalent assemblage of about two-thirds olivine, one-quarter low-Ca pyroxene, and about one-tenth garnet and clinopyroxene, would give a mean FeO/MnO near 70 using mean terrestrial values from Fig. 2. If the moon's *crust* contains little Ca-rich pyroxene and has a high ratio of olivine to low-Ca pyroxene, its bulk FeO/MnO ratio may be only a little less than the mean value of about 90 for Fig. 1. If the bulk moon

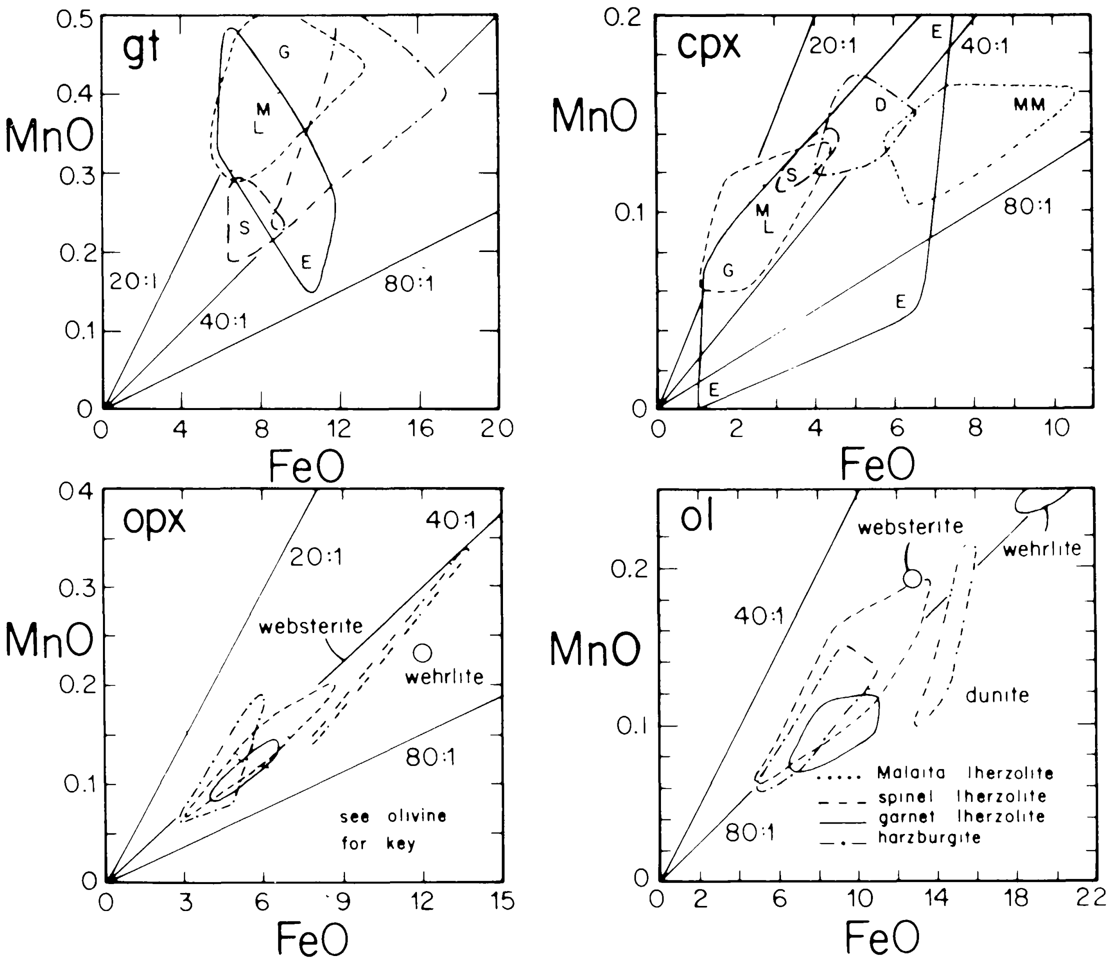

Fig. 2. Relationship between MnO and FeO (wt.%) for olivine, orthopyroxene, clinopyroxene and garnet from the earth's mantle. This diagram from Delaney *et al.* (1979a) is a summary of hundreds of electron microprobe analyses of the following samples. Contours show ranges for lherzolite inclusions from alnöite, Malaita, Solomon Islands; garnet lherzolite and harzburgite inclusions from kimberlites, southern Africa; spinel lherzolite inclusions in "kimberlites", south-western USA. Abbreviations in diagrams for garnet and clinopyroxene are: D discrete megacryst, probably phenocryst from a trapped magma; E eclogite xenolith from s. African kimberlites; G and S granular and sheared lherzolites from s. African kimberlites: ML and MM lherzolite and megacryst from Malaita, Solomon Islands. Some analyses are given in Delaney *et al.* (1979b,c; 1980) and Bishop *et al.* (1978), and many others are in various stages of incorporation into manuscripts.

consists of an olivine-rich mantle formed by gravity separation from a liquid which went on to generate a crust with a lower olivine content than the mantle, the FeO/MnO ratio of the whole moon should be greater than the value for the crust by a model-dependent amount, perhaps in the region of one-quarter. Hence the bulk Fe/Mn ratio of the moon might be greater than the value of 82 used in a Ganapathy-Anders type of model (Anders, 1977), and a value of 100±10 might be more plausible. Further speculation is futile except in the context of whole-moon models.

Comparison of FeO/MnO ratios of lunar samples with meteorites may be more profitable than comparison with the earth because crystal-liquid differentiation must have proceeded at relatively low pressure in the moon and parent bodies of meteorites with little complication from high-pressure phases. However, the effects of partitioning between minerals and liquids, and the uncertainties of petrologic models for meteorite parent bodies, must be considered. All existing analyses for olivines from eucrites, howardites, diogenites, mesosiderites, chassignites and ureilites give lower FeO/MnO than for lunar olivines (Hansen *et al.*, 1981, Fig. 4). Olivines from most pallasites also have low FeO/MnO (32–55; Delaney *et al.*, 1979a), but olivines from the Eagle Station pallasite and the Murchison C2 meteorite have ratios in the lunar range. For simple one-stage petrologic models for differentiation of meteorite parent bodies, it is possible to conclude that the part of the moon which differentiated to produce the observable crustal rocks was assembled from materials with higher FeO/MnO than most meteorite parent bodies: only the Eagle Station pallasite, the diogenites and the Murchison C2 meteorite have olivines with high enough FeO/MnO to be potential matches with the moon. In general, the moon is more refractory on the basis of the FeO/MnO ratio than most meteorites, as is consistent with other geochemical features such as the content of U and alkalies.

Calcium

For terrestrial olivines, Simkin and Smith (1970) found that those with greater than 0.14 wt.% CaO came from extrusive and hypabyssal rocks whereas those with less came from plutonic rocks. This distinction apparently carries over to lunar rocks (Steele and Smith, 1975, Fig. 1). Ricker and Osborn (1954) and Warner and Luth (1973) found that the Ca content of forsterite coexisting with monticellite increases with temperature. Duke (1976) found 0.50–0.58 wt.% CaO in olivines coexisting with Ca-rich pyroxene and liquid at 1150°C in the Na_2O-CaO-MgO-FeO-Al_2O_3-SiO_2 system. Experiments in the CaO-MgO-Al_2O_3-SiO_2 system showed that in olivine coexisting with orthopyroxene and clinopyroxene the Ca content increased with increasing temperature and decreasing pressure for the range 10–50 kb and 1000–1400°C (Finnerty and Boyd, 1978). For the low pressure range (<5 kb) of the lunar crust, the Ca content of olivine provides a potential thermometer providing rocks of appropriate composition are found. Unfortunately the olivine-orthopyroxene-clinopyroxene assemblage is very rare in lunar rocks, and is represented only by the 67667 feldspathic lherzolite (Hansen *et al.*, 1981, this volume) and the 60255,73α feldspathic microgabbro, for which temperatures of ~750°C and ~1000°C are obtained from Finnerty and Boyd (1978, Fig. 45) by linear extrapolation to ~2 kb. The two-pyroxene thermometer for the 67667 lherzolite gives 1055–1071°C, depending on the calibration, but there is a problem of obtaining reliable electron microprobe data for individual phases in the fine-scale intergrowth. For 60255,73α, the electron microprobe analyses (Table 2) yield two-pyroxene temperatures of 1090–1150°C (Wood and Banno, 1973) and 1140–1220°C (Wells, 1977). Although the Ca-in-olivine thermometer gives lower temperatures than the two-pyroxene thermometer, there are several possible systematic errors that must be studied thoroughly.

The mean values of 220 to 770 ppm CaO for all but one olivine in Table 2 correspond to the plutonic range of Simkin and Smith, and only the olivine from feldspathic microgabbro 60255,73β has a range (1590–1820; mean 1670) which overlaps into the volcanic and hypabyssal range of Simkin and Smith. Unfortunately there are no experimental determinations of the variation of Ca content of olivine coexisting with plagioclase, and such a study should be given high priority. A temperature estimate could be obtained from the Finnerty-Boyd data if the Ca content of olivine is assumed to have the same temperature dependence for equilibrium with either Ca-rich pyroxene or Ca-rich plagioclase. A second assumption is needed to extrapolate the data to lower temperature. The contours given by Finnerty and Boyd are essentially linear with respect to pressure and temperature, and for zero pressure extrapolate to zero CaO at ~600°C. An Arrhenius type of equation was calculated for the data pairs CaO 1728 ppm, 1000°C; 2155, 1100°C; 2617, 1200°C; 3078, 1300°C:

$$\ln(\mathrm{CaO}) = 10.506 - 3818(1/\mathrm{T})$$

where CaO is in ppmw and T is in Kelvin. This equation leads to lower temperatures than for the linear extrapolation by Finnerty and Boyd: e.g. 650°C for the 67667 feldspathic lherzolite instead of ~750°C. The lowest temperature is 400°C

Table 2. Analyses of low- and high-Ca pyroxene in 60255,73.

	low-Ca			high-Ca		
	1	2	3	4	5	6
SiO_2	53.9	55.1	55.7	53.6	52.3	53.2
TiO_2	na	na	0.50	.84	1.14	.95
Al_2O_3	1.88	1.15	1.67	1.71	1.77	1.71
Cr_2O_3	.78	.61	.73	.70	.75	.66
FeO	14.3	13.8	12.9	11.1	11.2	9.18
MgO	26.1	25.8	25.1	17.4	15.7	16.4
CaO	3.73	4.75	5.95	16.7	16.7	19.2
MnO	.22	.24	.37	.26	.24	.16
Σ	100.91	101.45	102.92	102.31	99.8	101.46

na = not analyzed.

for 15445,60 peridotite whose olivine contains only ~110 ppm CaO (Steele and Smith, 1975). A final problem with application of the Finnerty-Boyd data is lack of knowledge of the effect of Fe,Mg substitution in olivine on the Ca content. Because the boundary between plutonic and non-plutonic olivines seems to stay at ~0.14 wt.% CaO as the Mg/Fe ratio changes in terrestrial olivines from forsterite to fayalite, it will be assumed that the Mg/Fe ratio has little effect on the pressure-temperature control on Ca content.

Whatever the details, it is concluded that olivines in the coarse-grained lunar rocks have equilibrated to various sub-solidus blocking temperatures. Hence it is not feasible to deduce the Ca content of the parent liquid by use of the experimental partition data of Watson (1979) for coexisting olivine and liquid in the Na_2O-CaO-MgO-Al_2O_3-SiO_2 system, and by Duke (1976) for three compositions in the above system with added FeO. However, the Ca content of the liquid should be at least 20 times the Ca content of the olivine. Most data in the experimental study of Akella *et al.* (1976) indicate that Ca fractionates more strongly into olivine than basaltic liquid as the oxygen fugacity is reduced, but further measurements are desirable to test this conclusion before it is applied to interpretation of Ca content of lunar olivines.

Titanium

Fig. 3 suggests that there may be a weak positive correlation between the Ti contents of coexisting olivine and plagioclase in lunar rocks. The large spread of Ti concentrations found in olivines of some lunar rocks makes it difficult to

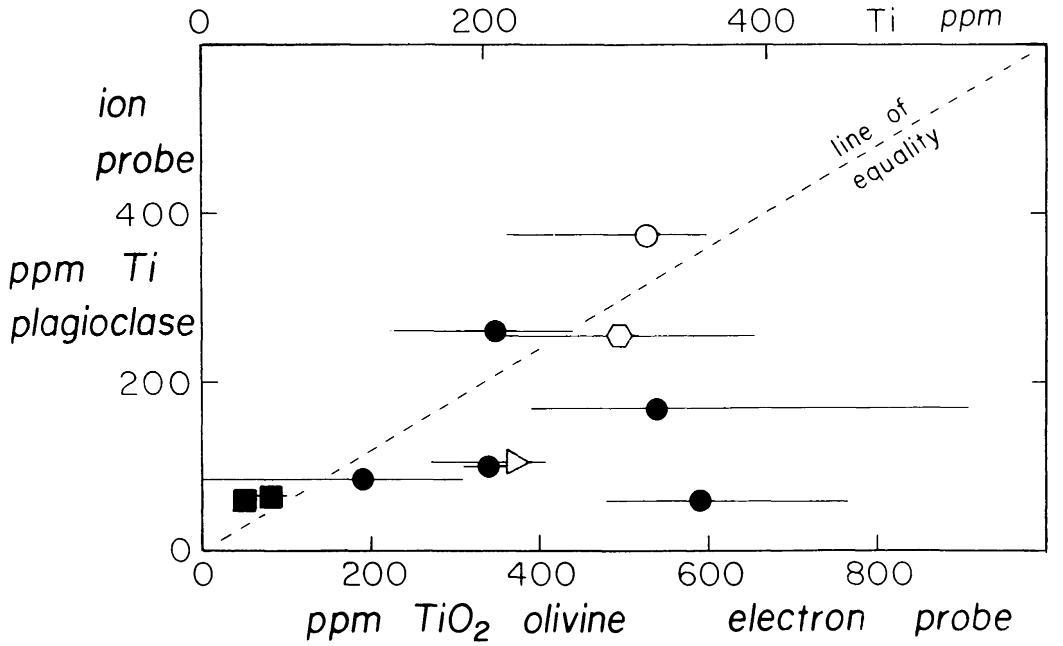

Fig. 3. Relationship between Ti concentrations in olivine and plagioclase. Data from Table 1 and Steele *et al.* (1981, this vol.). Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols given in Fig. 1.

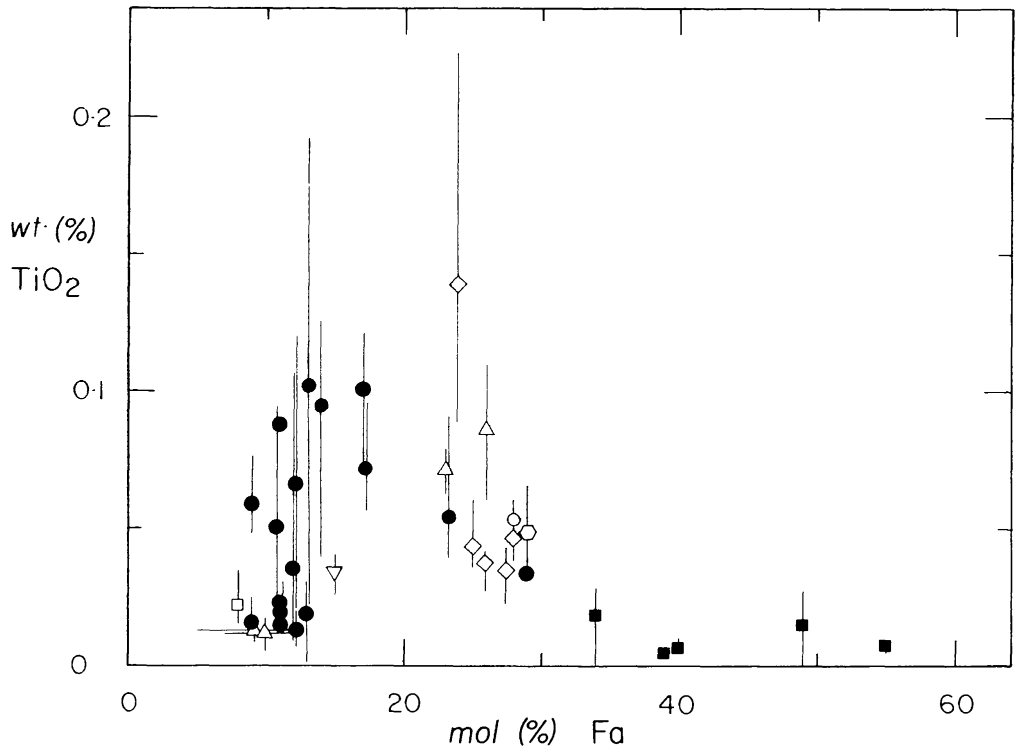

Fig. 4. Relationship between wt.% TiO_2 and mole % fayalite in olivines. Note large range of TiO_2 about mean value for some specimens. Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols given in Fig. 1.

determine how significant is the correlation, and many more data are needed especially of Ti in plagioclase for which only a few ion microprobe analyses have been made. The biggest deviations from the straight line for equal concentrations of Ti in olivine and plagioclase are for the 77135,121 and 73215,234 troctolite clasts, in which Ti favors olivine over plagioclase.

Iron-rich olivines from anorthosites have lower TiO_2 (50–190 ppm) than olivines from granulitic impactites (350–1890), some *mg*-rich plutonic rocks (110–590), and the three miscellaneous specimens (Table 1). During crystal-liquid fractionation of basaltic magmas, Ti tends to fractionate strongly and Fe/Mg weakly into basaltic liquid. The partitioning of Fe/Mg between olivine and liquid is affected only slightly by bulk composition and physical factors (Longhi *et al.*, 1978). All published data on the experimental partition of Ti between liquid and olivine (Irving, 1978, Table 1; Akella *et al.*, 1976) show a very strong preference for liquid, even when Ti-bearing spinel is crystallizing. Thus Akella *et al.* (1976) found a 22 to 47-fold enrichment of Ti in spinel over olivine and a 30 to 80-fold enrichment of Ti in liquid over olivine in experiments at one atmosphere and 1175–1300°C near the Fe-FeO buffer. A plot of TiO_2 vs. fayalite content of olivines (Fig. 4), however, does not show a simple positive correlation between titanium and iron. There is a large scatter, and the average of the band of data would give a curve with a maximum at about 0.1 wt.% TiO_2 and 20 mol % Fa. This figure is analogous to the plot of MgO vs. TiO_2 by Weiblen *et al.* (1974), and readers are referred there for a discussion of the effect of olivine, plagioclase and ilmenite crystallization on magma composition. There is no need to be concerned about the wide range of data for olivines from non-anorthositic specimens in view of the possibility, or indeed probability, of a diverse range of individual plutons in the lunar crust. However, the low Ti content of olivines from the Fe-rich anorthosites is yet another confirmation of the paradox that silicate minerals from anorthosites are richer in the magmatophile element Fe than minerals from *mg*-rich plutonic rocks and various breccias, even though they are poorer in other magmatophile elements. This paradox was first demonstrated for Na in plagioclase, and is now seen to be obeyed by a magmatophile element in olivine.

Because there is no correlation between TiO_2 and CaO, there is no reason to suspect that the Ti content of olivine decreases upon annealing to a subsolidus temperature. This contrasts with the strong positive correlation between Cr_2O_3 and CaO (next section).

Chromium

Experimental data summarized in Irving (1978, Fig. 6),and extended by Schreiber (1979) and Schreiber and Andrews (1980), demonstrated that Cr is partitioned approximately equally between olivine and liquid, with some change of the partition coefficient with oxygen fugacity depending on the temperature, bulk composition and range of oxygen fugacity (e.g., Schreiber and Haskin, 1976; McKay and Weill, 1976). The Cr content in a basaltic liquid is strongly controlled by the presence or otherwise of Cr-rich spinel as a crystallizing phase, and is also affected by crystallization of Cr-bearing pyroxene. On Earth, early Cr-rich minerals rapidly strip Cr away from the residual liquid in layered basaltic complexes, and there is a negative correlation between Cr and Fe in terrestrial orthopyroxene (Howie and Smith, 1966), and probably also for terrestrial olivine. There are no experimental data on the partition of Cr between olivine and other minerals at sub-solidus temperatures.

Fig. 5 shows that the Cr_2O_3 content tends to increase with the forsterite content of the lunar olivines from non-mare rocks as shown also in Smith (1974, Fig. 6). However, the spread of data is too large to be attributed to a simple crystal-chemical control. Because the highest Cr content is found for olivines in fragments with igneous texture (dashed line), a plot of Cr_2O_3 vs. CaO (Fig. 6) was made to test whether the chromium content was a function of the blocking temperature during sub-solidus cooling. Considering the diverse origin of the samples, and the large deviation of individual analyses about the mean value for a rock or clast, the positive correlation is reasonably good. The straight line is drawn merely as a guide, and has no statistical significance. Just as for other magmatophile elements in olivine and plagioclase in anorthosites, there is evidence that the anorthosites do not belong to the same chemical system as the other non-mare rocks. As a final test, Fig. 7 shows the relationship between TiO_2 and Cr_2O_3. Although there is a weak positive correlation, it is not as good as the correlation between calcium and chromium.

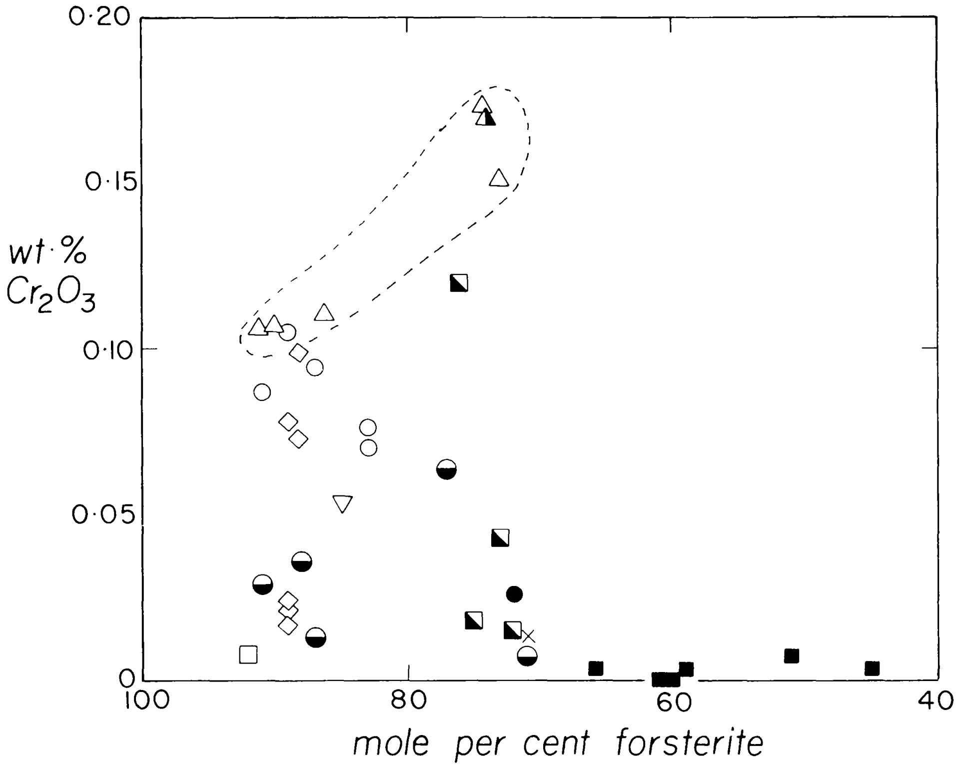

Fig. 5. Relationship between wt.% Cr_2O_3 and mole percent Fo in olivines. See tables for large ranges of Cr_2O_3 analyses. Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols given in Fig. 1.

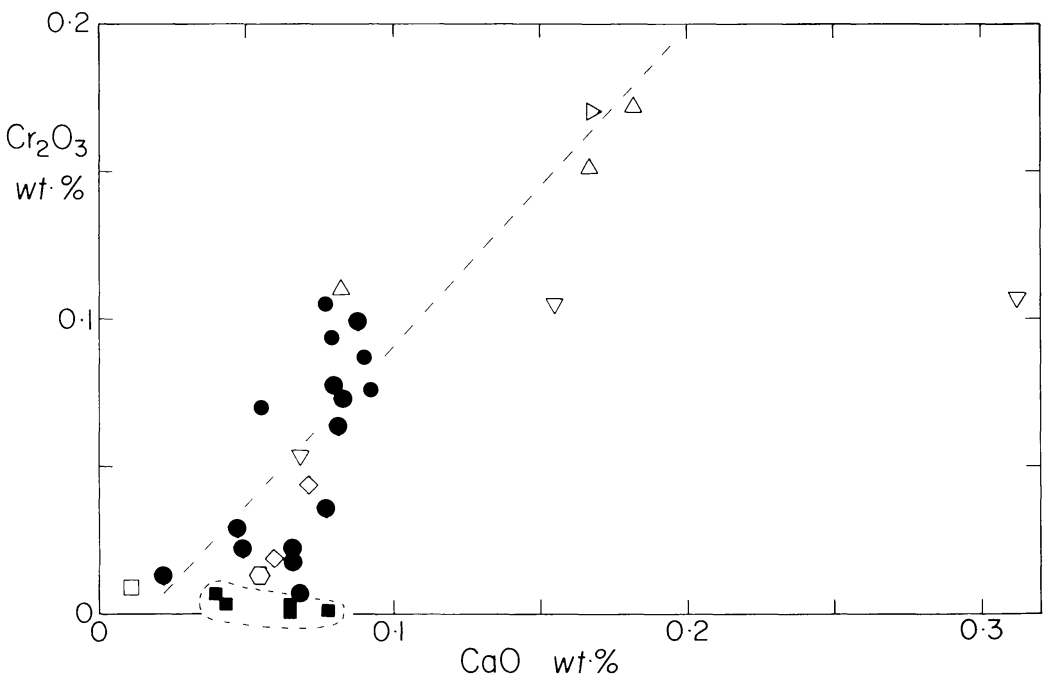

Fig. 6. Relationship between wt.% Cr_2O_3 and wt.% CaO. Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols given in Fig. 1.

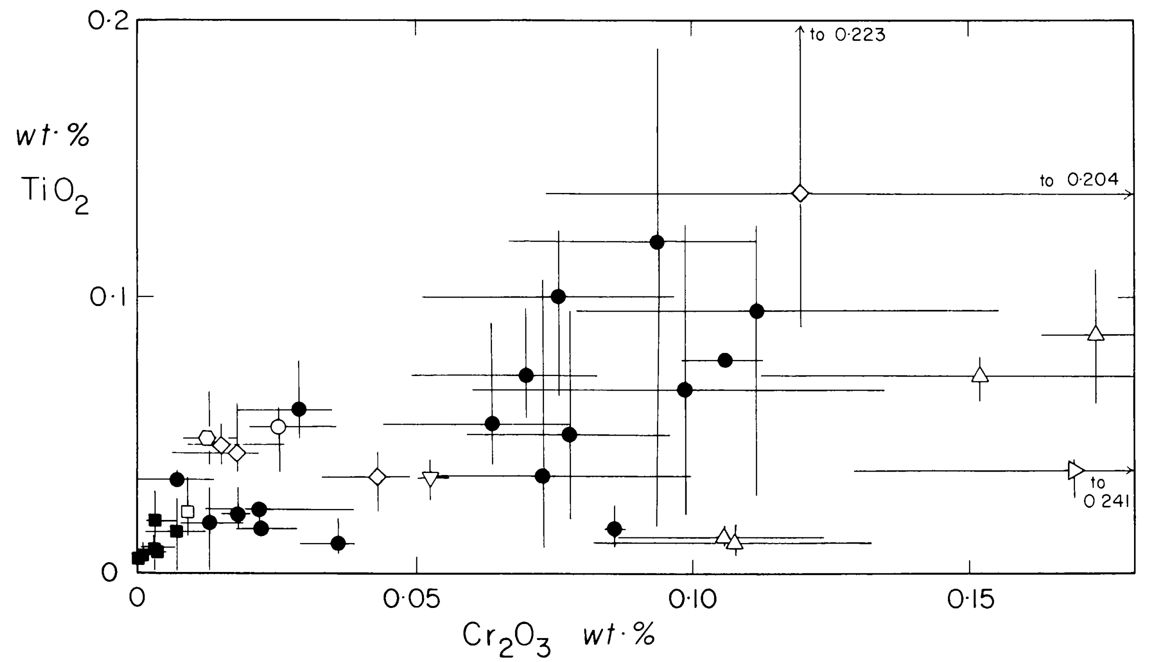

Fig. 7. Relationship between wt.% Cr_2O_3 and wt.% TiO_2. Solid squares—anorthosites; solid circles—high *mg* samples including olivine breccia, norite, troctolite; other symbols given in Fig. 1.

It is concluded tentatively that for the non-anorthositic specimens, the Cr content of the olivine depends strongly on the blocking temperature for equilibration with other minerals. Inclusions of Cr-bearing spinel have been reported in lunar olivines (Frondel, 1975), but a diffusion gradient of Cr in olivine around a spinel inclusion has not been reported. A thorough study is desirable to check for (i) evidence of a diffusion gradient between olivine and an adjacent Cr-rich silicate or oxide mineral, and (ii) evidence of Cr-rich inclusions in olivine in the present specimens. Zoning of Ca, Cr, Ti and other transition elements was found in olivine from the Springwater pallasite (Leitch *et al.*, 1979). Most terrestrial olivines contain little Cr, and become altered to hydroxylated minerals, but inclusions of spinel have been observed (Goode, 1974).

Other elements

Aluminum does not correlate with other elements, and the range of 0.015–0.04 wt.% Al_2O_3 lies inside the range 0.006–0.09 wt.% for all lunar olivines (Smith and Steele, 1976). All anorthositic olivines have low P_2O_5 (mostly <0.01 wt.%), whereas other olivines have a wide range from <0.01 to ~0.1 wt.%. These values compare with 0–0.02 wt.% in olivines from terrestrial lherzolites (Bishop *et al.*, 1978). Whereas most lunar olivines contain no detectable Na_2O, as for low-pressure olivines from the Earth, those from granulitic impactite 67746,1 contain an average of 0.03 wt.%. The only other known Na-containing olivines are from high-pressure peridotites from the Earth's mantle (Hervig, 1979) and diamonds (Hervig *et al.*, 1980). Nickel was not detected in the present specimens at the 100 ppm level, except for olivine in a troctolite clast (0.02 wt.%) and a granulitic impactite clast (0.03 wt.%), both from 73215. Steele and Smith (1975) found a range of 0–0.07 wt.% NiO in olivines from a variety of non-mare rocks.

CONCLUSIONS

The principal conclusion from the present study is that all the minor elements are very low in olivine from anorthosites, in spite of the high Fe content. Olivines from other non-mare rocks contain a higher and a relatively wide range of minor elements which suggests that these rocks crystallized from a different type of chemical and tectonic environment than the anorthosites. This conclusion matches that obtained by Steele *et al.* (1981) in an accompanying paper on the minor and trace elements of plagioclase in non-mare rocks. The calcium content of lunar olivine from non-mare rocks is tentatively explained by sub-solidus reaction with plagioclase, and an empirical temperature range down to 400°C for the 15445 peridotite is inferred very tentatively. The Cr content of olivine from non-anorthositic rock correlates positively with the Ca content, and is therefore believed to decrease with decreasing blocking temperature for metamorphic reaction with Cr-rich minerals. Olivines from anorthositic rocks have lower Cr than the olivines from other non-mare rocks, and the possibility that the former rocks crystallized from a magma with lower Cr than the latter rocks should be explored.

These tentative conclusions require checking by further experimental studies of the element distribution between olivine and other minerals at sub-solidus temperatures. Detailed studies are needed to check for diffusion gradients in those specimens that have not been shocked too severely.

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